Yiğit Sarp Avcı

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EDUCATION

Boğaziçi University — B.Sc. in Computer Engineering 09/2023 – Expected 06/2028

- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi; Regional Organizational Team Member at ESTIEM; Work and Travel USA participant; Student-Athlete (sailing, skiing & windsurfing).
- Volunteering: Volunteering Division Member at Industrial Engineering Club (BÜYAK); Educational Mentor to high school students.

Kabataş Boys' High School 09/2018 – 06/2023

- GPA: 3.78/4.00
- Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Clubs.
- Volunteering: Community Involvement Projects Club and TOÇEV (Tüvana Okuma İstekli Çocuk Eğitim Vakfı).

EXPERIENCE

DefineX — Backend Developer Intern 09/2026 – Present

- Developing internal data processing platforms and Model Context Protocol (MCP) integrations, streamlining operational workflows for corporate teams by automating enterprise data extraction and RESTful backend pipelines.
- Leveraging modern AI-assisted engineering workflows to accelerate enterprise backend feature delivery, implementing robust data validation schemas and optimizing API request throughput.

Akbank — Site Reliability Engineering Intern 07/2026 – 09/2026

- Maintained high availability for mission-critical core banking infrastructure by designing real-time observability dashboards and proactive alerting thresholds in Dynatrace and Grafana.
- Engaged in Linux systems engineering, server health diagnostics, and log analysis to accelerate incident detection and root-cause resolution across high-throughput production nodes.

Smash Mate — Software Engineering Intern 06/2026 – 07/2026

- Engineered player matchmaking, search discovery, and scoring microservices using Java (Quarkus), designing high-performance RESTful endpoints and core game-matching domain logic.
- Developed responsive mobile interfaces in React Native and authored comprehensive unit and integration test suites, delivering a fully functional, zero-defect pre-launch release.

TECHNICAL PROJECTS

- **Algorithmic Trading Simulator** — Quantitative engine executing Z-Score mean reversion on 2s cycles with volatility sizing; streams order books and P&L metrics via FastAPI WebSockets and Redis.
- **Financial Analytics Dashboard** — Personal finance platform computing multi-asset diversification scores and net cash flow; delivers context-aware advisory insights via OpenAI with rule fallbacks.
- **Restaurant Operations & Table Management SaaS** — Multi-tenant platform with slug-based isolation, interactive drag-and-drop floor plan builder (react-rnd), table merging, and waitstaff queues via Zustand and Zod.
- **Enterprise Operations & Knowledge Platform** — Workforce operations platform for daily activity audits, productivity benchmarking, and internal SOPs; features an Express 5 REST API, TypeORM, Recharts analytics, and SWR optimistic caching.
- **Linux IPC Game Subsystem** — 60 FPS modular game architecture decoupling input via non-blocking FIFOs (O_NONBLOCK), POSIX signals, lock-free ring buffers, and dlopen controller hot-swapping.
- **Cross-Platform & WebAssembly Game Lab** — C++/Qt6 2048 game compiled to WASM via Emscripten with an infinite undo stack (std::stack); and React Native Minesweeper mobile game featuring iterative BFS flood-fill and pure reducer state architecture.
- **Low-Level Systems & Architecture Lab** — Zero-libc x86-64 scheduler (raw syscalls), bare-metal RV32I navigator on Spike with linker script memory maps, C11 shell, and CoreMark ILP/compiler profiling.
- **Advanced Data Structures & Algorithms Suite** — $O(V + E)$ Tarjan's SCC graph analyzer, custom max-heaps with index tracking for $O(\log N)$ matching, a deterministic FSM engine, and an AST functional interpreter.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++, TypeScript, JavaScript, SQL, x86-64 Assembly, RISC-V Assembly
- **Technologies & Frameworks:** Next.js, React Native, Quarkus, FastAPI, Express, Tailwind CSS, Prisma, PostgreSQL, Redis, Docker, WebSockets, WebAssembly, Qt6
- **Developer Tools:** Git/GitHub, Linux/POSIX, AI-Assisted Development
- **Languages:** Turkish (Native), English (Advanced), German (Beginner)
- **Hobbies:** Wakeboarding, Kiteboarding, Surfing, Bouldering, Running, Traveling